

CAR-Net: A Deep Learning-Based Deformation Model for 3D/2D Coronary Artery Registration

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Abstract—Percutaneous coronary intervention is widely applied for the treatment of coronary artery disease under the guidance of X-ray coronary angiography (XCA) image. However, the projective nature of XCA causes the loss of 3D structural information, which hinders the intervention. This issue can be addressed by the deformable 3D/2D coronary artery registration technique, which fuses the pre-operative computed tomography angiography volume with the intra-operative XCA image. In this study, we propose a deep learning-based neural network for this task. The registration is conducted in a segment-by-segment manner. For each vessel segment pair, the centerlines that preserve topological information are decomposed into an origin tensor and a spherical coordinate shape tensor as network input through independent branches. Features of different modalities are fused and processed for predicting angular deflections, which is a special type of deformation field implying motion and length preservation constraints for vessel segments. The proposed method achieves an average error of 1.13 mm on the clinical dataset, which shows the potential to be applied in clinical practice.

Index Terms—3D/2D coronary artery registration, deep learning, centerline decomposition, angular deflections prediction.

I. INTRODUCTION

PERCUTANEOUS coronary intervention is routinely performed for treating coronary artery disease due to its ability to minimize physical trauma and reduce recovery time [1]. Currently, in clinical practice, the intervention is solely guided by X-ray coronary angiography (XCA) which provides real-time and multi-view images. However, the projective nature of XCA results in the loss of spatial information, which

creates ambiguities for the interpretation of original 3D vessel structures and further hampers the interventional procedure.

This issue can be addressed by a 3D/2D coronary artery registration technique, which fuses the vessel model extracted from the pre-operative 3D computed tomography angiography (CTA) volume with that extracted from the intra-operative 2D XCA image to achieve better guidance. However, due to the complexities of the coronary arteries and the big differences between the two image modalities, it's difficult to match the vascular structures. Besides, there are significant nonlinear deformations caused by the heartbeat and the respiration existing in each vessel segment, which makes the 3D/2D coronary artery registration task challenging. Some literature [2]–[16] has focused on this task and made certain achievements. However, there are still limitations. First, the existing methods adopt the traditional iterative optimization idea that is time-consuming and generate the deformation field with common algorithms such as the thin-plate spline (TPS) [6], [7], [14] and the statistical shape model [11], [16], whose accuracies could be further improved. Second, the existing methods usually model the deformation field with global Euclidean displacement [6], [7]. It has a large range and requires explicit constraints for the length of the vessel segment, which hampers the algorithm convergence.

To overcome these limitations, in this study, we propose a specially designed registration model: Coronary Artery Registration Network (CAR-Net). The deep learning-based registration theory [17]–[20] is introduced to overcome the first limitation, which generates the deformation field by a neural network in an end-to-end manner without a complex optimization process at the inference time. Different from the existing intensity-based 3D/2D registration networks [21]–[25], CAR-Net is feature-based, which takes vessel centerlines that preserve the topological information of coronary trees as input. Since it's difficult to convert different coronary trees into a unified data structure for network input, CAR-Net splits a coronary tree into a series of vessel segments following the topological structure and registers them one by one. Specifically, for reducing the intra-variance of centerline data, each input segment centerline is decomposed into an origin tensor and a spherical coordinate shape tensor that are processed by two independent branches. To overcome the second limitation, CAR-Net models the deformation field with local angular deflection instead of global Euclidean displacement. Angular deflection denotes the spherical coordinate variation of

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a 3D centerline point relative to its preceding point, which implies motion and length preservation constraints for each vessel segment and reduces the number of parameters to be predicted.

After CAR-Net is defined, we evaluate its performance with both synthetic deformation data that is generated to increase data diversity and real clinical data. By fine-tuning a synthetic deformation data pre-trained network, the performance of CAR-Net on clinical data can be improved, achieving a faster convergence speed and a better solution.

A. Contributions

The main contribution of this study is that we propose a novel coronary artery registration model CAR-Net, which conducts the registration in a segment-by-segment manner, and innovatively introduces the input centerline decomposition and the angular deflections prediction. And to the best of our knowledge, this is the first work that conducts deformable 3D/2D coronary artery registration with deep learning.

B. Related Works

1) 3D/2D Coronary Artery Registration (Non-Learning-Based): The existing coronary artery registration methods can generally be categorized into intensity-based and feature-based. Intensity-based methods [3], [4] measure similarity with gray intensity difference between digitally reconstructed radiograph generated from CTA volume and XCA image. However, since vessel structures are sparse in both modalities, registration accuracy is damaged by large and insignificant backgrounds.

Feature-based methods [6]–[16] achieve better performance by measuring similarity with coronary centerlines and could be classified into three categories according to the rules of matching projected CTA points with XCA points. Methods of the first category restrict the matching rule to a simple one-to-one manner [6], [7]. Euclidean distance often serves as the optimization target with designed constraints for vessel length, smoothness of displacement field, etc. However, such methods are sensitive to noise and outliers, and the optimization process is vulnerable to local minima due to the one-to-one restriction. Methods of the second category relax the matching restriction from one-to-one to one-to-many by establishing a probability framework with variants of the coherent point drifts (CPD) algorithm [8]. Kang *et al.* [9] were the first to introduce the CPD algorithm to 3D/2D point registration, which regarded the 3D point sets as the Gaussian mixture model that generated the 2D point set. And registration was performed by maximum likelihood estimation. Jian *et al.* [10] improved the CPD algorithm by regarding both point sets as Gaussian mixture models and optimizing the overlap L2 distance between the two models. Baka *et al.* [11] further updated the distance metric by incorporating the orientation information of centerline points. Methods of the third category introduce topological information to registration by treating the vessel segments and bifurcations as a graph. Benseghir *et al.* [12] extended the iterative closet point algorithm to the iterative closet curve algorithm by using the vessel segment as the pairing

element instead of point and measured the similarity of two centerline curves with Fréchet distance. On this basis, the authors [13] imposed consistency to the bifurcation points to ensure a one-to-one matching of the centerline curves. Liu *et al.* [14] further improved this matching strategy from one-to-one node to many-to-many nodes under the constraints of tree topology. Zhu *et al.* [15], [16] adopted a coarse-to-fine strategy, which matched the bifurcation nodes with a redundant graph matching method in the coarse stage and matched the vessel points by assembling the edge-to-edge results in the fine stage.

2) 3D/2D Medical Image Registration (Learning-Based):

Although most of the existing deep learning-based medical image registration methods handle registration across the same dimension, more works begin to focus on 3D/2D registration. Miao *et al.* [21] might be the first to propose a convolutional neural network-based regression model for transformation parameter estimation. However, the efficiency was damaged since features were extracted from small image patches and the method was only applicable to rigid object transformation. The authors further applied reinforcement learning to search for the optimal updating direction of the transformation parameters with multiple agents [22], which showed high robustness and efficiency. Liao *et al.* [23] conducted registration via a point-of-interest network, which was designed for landmark tracking and triangulation. The transformation parameters were estimated by a triangulation layer after the points were matched. Guan *et al.* [24] paid attention to the data problem, which shared the image characteristics of different patients by transfer learning to shorten the training time and reduce the number of images needed for training. The above-mentioned works all applied the digitally reconstructed radiograph as network input with no gradient connection to 3D space. To solve this problem, Gao *et al.* [25] were the first to propose a novel projective spatial transformer module. It enabled end-to-end learning of image feature extraction and projection, which can be optimized effectively using conventional gradient descent.

II. METHODS

The overall framework of the proposed coronary artery registration method is shown in Fig.1, which consists of the pre-processing module and the CAR-Net registration module. The following sub-sections will describe them in detail.

A. Pre-Processing Module

To extract the vessel centerlines and build correspondence across the two image modalities, pre-processings are required.

The CTA volume segmentation is conducted by a modified V-net [26] that processes the volume directly with volumetric instead of slice-wise convolutions. Then, the vessel centerlines are extracted with morphological operations. Considering that the centerlines are made up of continuous voxels in the volume space, the bifurcation points are recognized by searching for the voxels with more than three voxels in its neighborhood. Based on these bifurcations, the original centerline tree could

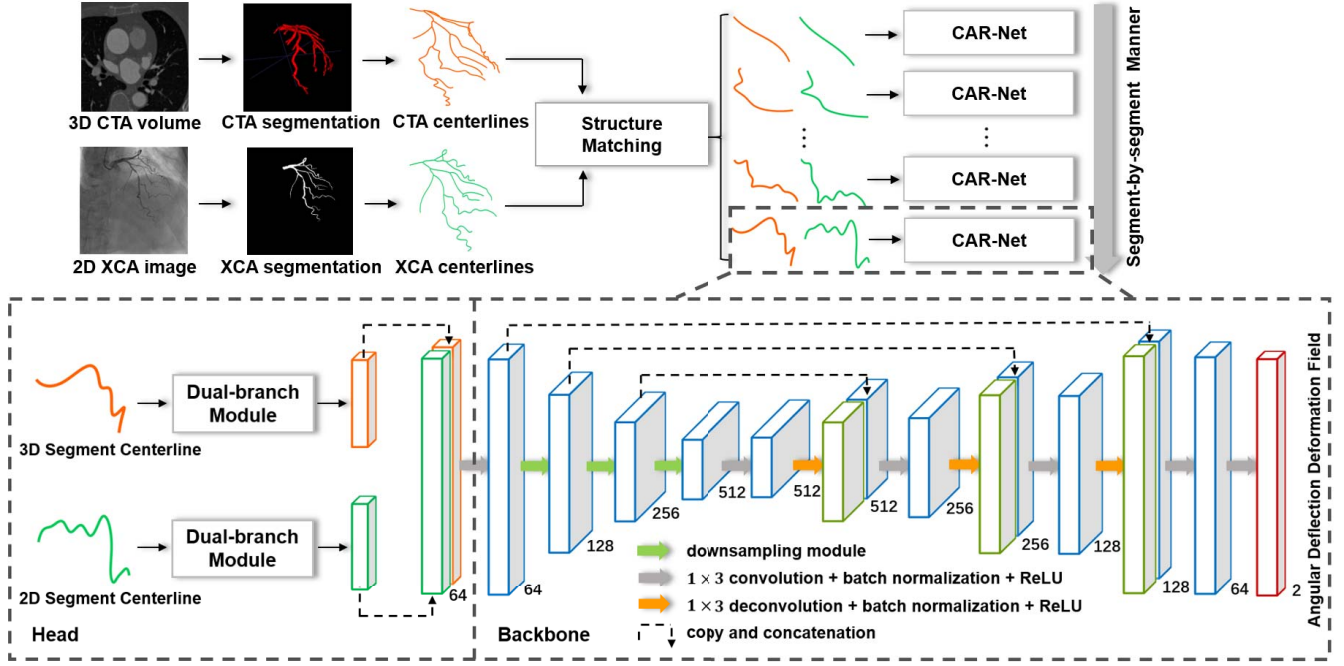


Fig. 1. The overall registration framework. First, the input vessel centerlines are generated with a series of pre-processing steps. Then, the proposed CAR-Net conducts registration in a segment-by-segment manner with centerline decomposition (dual-branch module) and angular deflections prediction.

be split into a series of centerline segments. The topological information of the centerline tree, which indicates the connections of the centerline segments, is essential for the subsequent segment-by-segment registration and is manually labeled (manual interaction could be applied to process the pre-operative CTA volume since the registration task is intra-operative). The XCA image segmentation is conducted by a weakly-supervised learning framework [27] that learns from noisy pseudo labels generated by vessel enhancement and involves minor manual corrections at the training time. The centerline extraction and the bifurcations recognition are conducted with similar methods previously applied to the CTA volume. However, there are fake bifurcations existing in the XCA images due to vessel overlapping caused by projection. Therefore, connectedness clustering is adopted to distinguish the fake bifurcations as in [14].

Fig. 2 shows a C-arm model, where the X-rays produced by the point source pass through the cardiac structures, reach the detector plane and generate an XCA image. This natural projection process is simulated mathematically to project the 3D CTA centerlines to the 2D XCA image plane for registration, where the most important step is to transform the centerlines from the CTA volume coordinate system to the X-ray source coordinate system. The key parameters, including the spatial angles α and β that determine the orientation of the detector plane, the distance between the X-ray source and the patient d_{xc} , and the distance between the X-ray source and the detector plane d_{xd} are recorded in the DICOM head file. First, the position of the X-ray source $p = (p_x, p_y, p_z)$ in the CTA volume coordinate system is computed. Since p_x , p_y and p_z are the projections of d_{xc} on the three axes and the

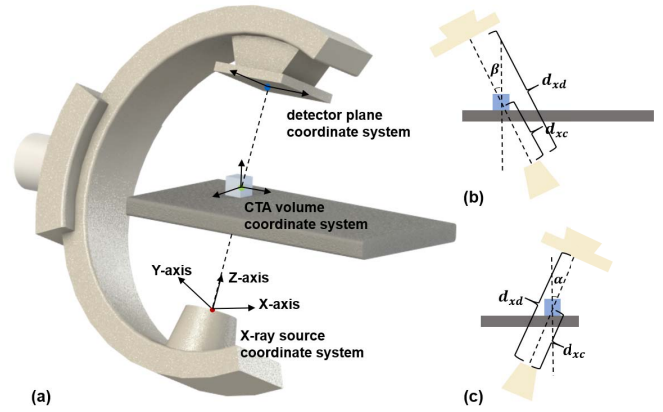


Fig. 2. The projection model for generating XCA image. (a) is the C-arm model. (b) β and (c) α are two spatial angles that determine the orientation of the detector plane towards the patient. d_{xc} is the distance between the X-ray source and the patient. d_{xd} is the distance between the X-ray source and the detector plane.

two projection angles α and β are given, p is computed by solving the equations below:

$$\begin{cases} p_x^2 + p_y^2 + p_z^2 = d_{xc}^2 \\ -p_x/p_y = \tan \alpha \\ -p_z/p_y = \tan \beta \end{cases} \quad (1)$$

The solution is:

$$p = (-d_{xc} \tan \alpha / p_{base}, d_{xc} / p_{base}, -d_{xc} \tan \beta / p_{base}) \quad (2)$$

where $p_{base} = \sqrt{1 + \tan^2 \alpha + \tan^2 \beta}$.

Then, the unit vectors x_{axis} , y_{axis} and z_{axis} of the coordinate axes of the X-ray source coordinate system are computed. z_{axis} is first computed since it points to the origin of the CTA volume coordinate system:

$$z_{axis} = -p/d_{xc} \quad (3)$$

According to the C-arm geometry, x_{axis} is parallel to the XY plane of the CTA volume coordinate system, which indicates that the z value of x_{axis} is 0. Since the inner product of x_{axis} and z_{axis} is zero and the norm of x_{axis} is 1, two equations can be defined to compute x_{axis} and the solution is:

$$x_{axis} = \left[p_y / \sqrt{p_x^2 + p_y^2}, -p_x / \sqrt{p_x^2 + p_y^2}, 0 \right] \quad (4)$$

Since y_{axis} is perpendicular to x_{axis} and z_{axis} and its norm is 1, three equations can be defined to compute y_{axis} and the solution is:

$$y_{axis} = \left[-p_x p_z / y_{base}, -p_y p_z / y_{base}, -p_z p_z / y_{base} \right] \quad (5)$$

where $y_{base} = \sqrt{(p_x^2 + p_y^2)^2 + p_x^2 p_z^2 + p_y^2 p_z^2}$.

Based on the computations above, a centerline point $v = (v_x, v_y, v_z)$ defined in the CTA volume coordinate system can be transformed to $r = (r_x, r_y, r_z)$ defined in the X-ray source coordinate system with:

$$r = ((v_x - p_x) * x_{axis}, (v_y - p_y) * y_{axis}, (v_z - p_z) * z_{axis}) \quad (6)$$

And the transformed 3D centerline point r can be projected to r' in 2D detector plane coordinate system with distance d_{xd} following the principle of similar triangle:

$$r' = [r_x d_{xd} / r_z, r_y d_{xd} / r_z] \quad (7)$$

After the coordinates of CTA centerlines are transformed from the CTA volume system to the X-ray source system, correspondence is established between the vessel segments across the two modalities. First, the root points are recognized by selecting the bifurcations with the maximum radius in the original coronary trees. A rigid displacement is conducted as a pre-alignment step to align the roots of the two modalities. Then, the vessel matching algorithm [14] is applied, which considers the vascular trees of the two modalities as two directed acyclic graphs and the bifurcations as nodes. And the structure matching problem is transformed into a many-to-many nodes matching problem with topological consistency constraint. In other words, the edge connected between two adjacent nodes in 3D will be matched with complete directed edges between the corresponding two nodes in 2D.

B. CAR-Net

After all the pre-processing procedures, the vessel segment centerlines are extracted and matched. Then, the registration is conducted in a segment-by-segment manner and a proximal to distal order based on the topological connection information of the coronary tree as shown in Fig.1. When a vessel segment is registered, its subsequent segments first move to the new positions to ensure the continuities at the bifurcation point

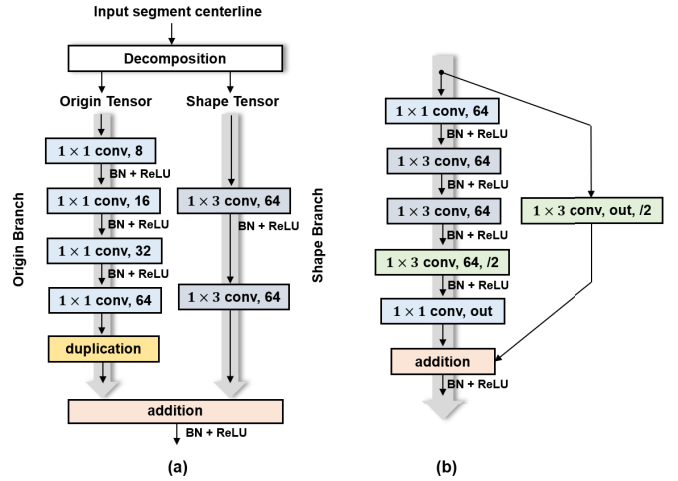


Fig. 3. The structures of the proposed (a) dual-branch module and (b) downsampling module. 'BN', 'ReLU', and 'out' denote batch normalization, Rectified Linear Unit, and output channel number. '/2' denotes stride=2.

and then be registered. The overall registration results are obtained by summarizing the registration result of each vessel segment generated by the proposed CAR-Net. Let $C_{3D} = [m_0, m_1, \dots, m_n] \in R^{3 \times n}$ denote a transformed CTA segment centerline, where $m_n = [a_{nx}, a_{ny}, a_{nz}]^T$ denotes a 3D point. $C_{2D} = [t_0, t_1, \dots, t_n] \in R^{2 \times n}$ denotes a raw XCA segment centerline, where $t_n = [b_{nx}, b_{ny}]^T$ denotes a 2D point. n is the point number. The proposed CAR-Net models a function $G_\theta(C_{3D}, C_{2D}) = F$, where θ and F denote the network parameters and the desired deformation field, respectively.

The most common form of the neural network input is the raw centerline data, just as applied in the existing intensity-based registration frameworks. However, at the head of CAR-Net, a novel input centerline decomposition operation is adopted, which aims to alleviate the intra-variance in centerline data and accelerate the convergence of the training process (detailed analysis is presented in section IV). The centerlines of the two modalities are fed into two independent branches and at each branch, the raw centerline is processed by the dual-branch module as shown in Fig.3(a). This module decomposes the input centerline into an origin tensor and a shape tensor and extracts their features by two smaller branches. In the proposed method, the shape tensor is defined with spherical coordinates. Taking the 3D centerline input C_{3D} as an example. $C_{3D} = [m_0, m_1, \dots, m_n]$ is redefined as $C_{3D} = [m_0, s_1, \dots, s_n]$ and decomposed into origin tensor $CO_{3D} = [m_0]$ with size 3×1 and shape tensor $CS_{3D} = [s_1, \dots, s_n]$ with size $3 \times (n - 1)$. m_0 denotes the Euclidean coordinate of the origin point and $s_n = [l_n, \theta_n, \phi_n]^T$ denotes the spherical coordinate of the n_{th} point relative to its preceding point. The original Euclidean coordinate of point m_n can be restored with:

$$m_n = m_0 + \sum_{i=1}^n \begin{bmatrix} l_i \sin \theta_i \sin \phi_i \\ l_i \sin \theta_i \cos \phi_i \\ l_i \cos \theta_i \end{bmatrix} \quad (8)$$

And as an alternative of the proposed spherical coordinate shape decomposition, we could also consider define the shape tensor with common Euclidean coordinates:

$CS_{3D} = [d_1, \dots, d_n]$, where $d_n = m_n - m_{n-1}$. After decomposition, the two tensors are processed by the proposed dual-branch module. The origin branch takes CO_{3D} as input and comprises of four one-dimensional convolutional layers with kernel size 1 and kernel number $\{8, 16, 32, 64\}$, which outputs an origin feature with size 64×1 . The shape branch takes CS_{3D} as input and has two convolutional layers with kernel size 3 and kernel number 64, which outputs a shape feature with size $64 \times (n - 1)$. The origin feature tensor is replicated along its last dimension for $(n - 1)$ times and added with the shape feature tensor, which generates the final 3D centerline feature with size $64 \times (n - 1)$. Simultaneously, the 2D segment centerline is processed by another dual-branch module with identical structures, which also outputs feature with size $64 \times (n - 1)$. Finally, the features of the two modalities are aggregated by concatenating along the first dimension for subsequent deformation field prediction.

The low-level features extracted by the head are further processed by the backbone of CAR-Net, which is a U-net [28] structure comprising of an encoder with four downsampling blocks and a decoder with four upsampling blocks. The downsampling block is a residual block with skip connection [29], shown in Fig.3(b). The channel number of the tensor is compressed to 64 for computation reduction by the first layer and expanded to the required size by the last layer with kernel size 1. While extracting features, the middle layers with kernel size 3 and kernel number 64 compress the tensor length to half with stride 2. The shortcut layer fuses the input with output by addition. The upsampling block is a transposed convolutional layer with kernel size 3 and kernel number 64 which restores the original length of the input tensor. Its output is fused with that of the equal-level downsampling block, generating features with richer semantic information. Finally, the output layer transforms the features into the desired deformation field for registration.

The existing registration algorithms usually predict global Euclidean displacement as a deformation field. This could be improved with local Euclidean displacement that records the displacement of a certain point to its preceding point since it has a smaller value and is easier to control the smoothness. Different from these common options, we propose a new way by outputting the local angular deflection of each 3D centerline point relative to its preceding point in a spherical coordinate system as the deformation field. This representation form comprises implicit motion and length preservation constraints for vessel segments and reduces the number of parameters to be predicted. The new centerline can be constructed with the input tensors and the output deformation field $F = [\Delta s_1, \dots, \Delta s_n]$ ($\Delta s_n = [\Delta \theta_n, \Delta \theta_n]^T$) with:

$$\begin{aligned} m'_n &= m_0 + \begin{bmatrix} x_1 & \dots & x_n \\ y_1 & \dots & y_n \\ z_1 & \dots & z_n \end{bmatrix} \begin{bmatrix} 1 & \dots & 1 & 1 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \dots & 1 & 1 \\ 0 & \dots & 0 & 1 \end{bmatrix} \\ &= m_0 + \begin{bmatrix} x_1 & x_1 + x_2 & \dots & \sum_{i=1}^n x_i \\ y_1 & y_1 + y_2 & \dots & \sum_{i=1}^n y_i \\ z_1 & z_1 + z_2 & \dots & \sum_{i=1}^n z_i \end{bmatrix} \quad (9) \end{aligned}$$

where m'_n is the new coordinate after deformation, $x_n = l_i * \sin(\theta_i + \Delta \theta_i) * \sin(\theta_i + \Delta \theta_i)$, $y_n = l_i * \sin(\theta_i + \Delta \theta_i) * \cos(\theta_i + \Delta \theta_i)$, $z_n = l_i * \cos(\theta_i + \Delta \theta_i)$ and $*$ denotes element-wise multiplication.

As introduced above, the input shape tensor could also be defined in the Euclidean coordinate system or be replaced with raw centerline data. And the deformation field could also be defined with global or local Euclidean displacement. Therefore, several variants of the proposed method could be generated by combining these alternatives at the input stage and the output stage. Comparative experiments are conducted in section III-C and III-E to show the superiority of the proposed method.

C. Loss Functions and Evaluation Metrics

During the network training process, after the deformed 3D centerlines are constructed based on the output deformation field and projected to the 2D XCA image plane with (7), a normalized soft dynamic time warping (soft-DTW) distance [30] is applied to measure the similarity between the projected moving curve $pro(C_{3D})$ and the target curve C_{2D} . Soft-DTW (denoted by DTW_γ and γ is the smoothing parameter) searches for a soft-minimum by considering all the possible alignments between two sequences. Both its high computation efficiency and differentiability are essential for neural network training. It is expressed as:

$$\begin{aligned} DTW_\gamma(pro(C_{3D}), C_{2D}) \\ = -\gamma \log\left(\sum_{p \in set(P)} e^{-(p, d(pro(C_{3D}), C_{2D}))/\gamma}\right) \quad (10) \end{aligned}$$

where $set(P)$ denotes all the possible paths, $\langle \rangle$ denotes inner product, $d(\cdot)$ denotes the Euclidean distance. Since the value of soft-DTW might be negative, which is unacceptable as a loss function for the neural network training, a normalization operation is further conducted to ensure its non-negativity:

$$\begin{aligned} L_{dtw}(pro(C_{3D}), C_{2D}) \\ = DTW_\gamma(pro(C_{3D}), C_{2D}) \\ - (DTW_\gamma(pro(C_{3D}), pro(C_{3D})) \\ + DTW_\gamma(C_{2D}, C_{2D}))/2 \quad (11) \end{aligned}$$

Besides, a smoothness constraint is also imposed on the output deformation field. A diffusion regularization on the spatial gradients of the angular deflections is used to encourage a physically realistic deformation field F :

$$L_{smooth}(F) = \sum (\|\nabla F_\theta\|^2 + \|\nabla F_\phi\|^2) \quad (12)$$

Therefore, the overall training loss L is expressed as:

$$\begin{aligned} L(C_{3D}, C_{2D}, F) \\ = L_{dtw}(pro(C_{3D}), C_{2D}) + aL_{smooth}(F) \quad (13) \end{aligned}$$

where a is a regularization parameter. Note that when predicting Euclidean displacement as in section III, an explicit constraint L_{length} for controlling vessel segment length should be added to (13):

$$L_{length} = \sum_{i=1}^n (\|m_i - m_{i-1}\|^2 - \|m'_i - m'_{i-1}\|^2) \quad (14)$$

Therefore, the current loss function is:

$$\begin{aligned} L(C_{3D}, C_{2D}, F) \\ = L_{dTw}(pro(C_{3D}), C_{2D}, F) \\ + aL_{smooth}(F) + bL_{length}(C_{3D}) \end{aligned} \quad (15)$$

where b is another regularization parameter.

After the training process is finished, the mean projection distance (mPD) [31] is used as the metric to measure the registration errors on the test set:

$$mPD = \sum_{i=1}^n \|pro(m_i) - cor(m_i)\| / n \quad (16)$$

where $cor(m_i)$ denotes the 2D corresponding point of m_i .

III. EXPERIMENTS AND RESULTS

A. Dataset

The clinical data were collected from 56 patients at the Department of Cardiothoracic Surgery, Peking Union Medical College Hospital of China. For each patient, a pre-operative CTA volume and about 5 or 6 intra-operative XCA sequences were obtained. The CTA volumes were acquired on the Siemens SOMATOM Definition Flash CT platform. The number of slices ranges from 210 to 380 and the resolution of each slice is 512×512 . The voxel size is $0.35 \times 0.35 \times 0.75$ (unit: mm). The XCA sequences were acquired on Philips INTEGRIS Allura Flat Detector. The frame rate is 15 fps. The number of frames ranges from 40 to 75 and the resolution of each frame is 512×512 . The pixel size is 0.37×0.37 (unit: mm). Data of 46 patients were randomly sampled as the training set and data of the remaining 10 patients were used as the test set. For each XCA sequence, only the middle contrast agent-filled frames are useful since they show the complete coronary artery structures. Considering that these frames are similar, one frame is selected to constitute a data pair with corresponding CTA volume. In total, there are 240 data pairs in the clinical training set and 54 data pairs in the clinical test set.

Empirically speaking, the size of our dataset is relatively small compared with the publicly available medical datasets that are usually collected from hundreds or even thousands of patients. Since some literature [32]–[34] has demonstrated that pre-trained models are beneficial for medical image analysis due to the increased data diversity, we also build a synthetic deformation dataset to pre-train the proposed CAR-Net. And the method for the dataset generation is based on [35]. For a CTA centerline segment, the spherical coordinates of the n_{th} skeleton point relative to its preceding point $s_n = [l_n, \theta_n, \phi_n]^T$ is first calculated. Then, the synthetic deformation field $\Delta s = [\Delta s_1, \dots, \Delta s_n]$ ($\Delta s_n = [\Delta \theta_n, \Delta \phi_n]^T$) that consists of two series: $\Delta \theta = [\Delta \theta_1, \dots, \Delta \theta_n]$ and $\Delta \phi = [\Delta \phi_1, \dots, \Delta \phi_n]$ are generated. The method in [35] generates the deformations by sampling from a gaussian distribution and further smooths the deformed shape to make it physically meaningful. We improve this by sampling from two sine distributions continuously, which implies smoothness and length preservations for each vessel segment. Finally, a deformed CTA segment centerline could be generated based on (9). The deformed CTA centerline tree is generated by connecting all the deformed CTA

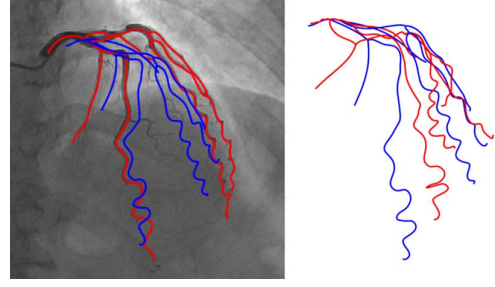


Fig. 4. Clinical data (left) and synthetic deformation data (right). Blue and red denote the original 3D coronary artery structure and the target 2D coronary artery structures, respectively.

TABLE I

THE COMMONLY USED PROJECTION ANGLE SETTINGS (α, β) FOR OBSERVING LCA AND RCA, RESPECTIVELY

artery	projection angle settings (α, β) ($^\circ$)
LCA	(-30, 25), (0, 25), (35, 25), (-30, -25), (0, -30), (45, -35)
RCA	(40, 0), (0, 25), (-30, 0)

segments following the original topology. With (1) ~ (7) and the projection angle parameter (α, β), the 3D centerline tree could be projected to the 2D detector plane. Note that there are some commonly used (α, β) for observing left coronary arteries (LCA) and right coronary arteries (RCA), respectively, as shown in Table I. Since deviations usually exist when setting (α, β) in clinical practice due to manual operation, a deviation from $(-10^\circ, 10^\circ)$ is also introduced when we apply (α, β) for projection. The synthetic deformation training set consists of 4968 coronary tree samples with 12 samples generated from each patient in the clinical training set under each projection angle setting. The synthetic deformation test set consists of 360 coronary tree samples with 4 samples generated from each patient in the clinical test set under each projection angle setting. Visualization of the clinical data and the synthetic deformation data are presented in Fig.4.

B. Implementation Details

The CAR-Net is implemented based on PyTorch framework with software environment: PyTorch 1.10.1, CUDA 9.0.176, cuDNN 7.0.3, Python 3.6.7, and operating system ubuntu 20.04 LTS and hardware environment: a 4-core 2.4 GHz Intel Xeon E5-2630 v3 processor and a single Nvidia Titan X Pascal GPU. The total number of parameters in CAR-Net is 1.31M. Based on the experimental settings above, the inference time is 0.006s. Considering that a coronary artery sample usually has several vessel segments, it takes about 0.05s to conduct a complete registration. The codes of CAR-Net are available at <https://github.com/GoldExperienceRequiemWW/CAR-Net>.

Each centerline segment is interpolated to 350 points ($n = 350$) with B-spline. When training with synthetic deformation data, the network weights are initialized with standard Gaussian distribution. The batch size is 512. The number of training epoch is 186. The learning rate is 0.1 initially and

TABLE II

THE MEAN PROJECTION DISTANCE (MM) $\pm$ STANDARD DEVIATION OF DIFFERENT METHODS ON SYNTHETIC DEFORMATION TEST SET AND THE P-VALUES OF CORRESPONDING T-TESTS. IN INPUT, 'RAW': RAW DATA; 'SPHERICAL': SPHERICAL COORDINATE SHAPE DECOMPOSITION; 'EUCLIDEAN': EUCLIDEAN COORDINATE SHAPE DECOMPOSITION. IN DEFORMATION, 'EUCLIDEAN': LOCAL EUCLIDEAN DISPLACEMENT FIELD; 'ANGULAR': ANGULAR DEFLECTION FIELD

method	input			deformation		mPD (mm)	p-value
	spherical	Euclidean	raw	Euclidean	angular		
proposed	✓				✓	0.80 ± 0.64	/
1 st variant		✓			✓	0.94 ± 0.62	1.35e-02
2 nd variant	✓			✓		1.54 ± 1.15	1.60e-24
3 rd variant		✓		✓		1.68 ± 1.15	3.49e-23
4 th variant			✓		✓	3.75 ± 2.23	1.84e-59
5 th variant			✓	✓		5.93 ± 3.68	3.23e-65

decays to 0.01 at the 48th epoch and 0.001 at the 135th epoch. When training with clinical data, a network pre-trained with synthetic deformation data is applied as the initial model. The batch size is 512. The number of training epoch is 300. The learning rate is 0.1 initially and decays to 0.01 at the 90th epoch and 0.001 at the 240th epoch. L2 regularization ($\lambda = 10^{-4}$), Adam optimizer, and batch normalization are applied at both synthetic deformation dataset experiment and clinical dataset experiment. The regularization parameter a of the smoothness preservation loss is 10. The regularization parameter b of the smoothness preservation loss is 10. Note that these common parameters are selected mainly based on experience. And mPD is used in the loss function instead of soft-DTW when experimenting with synthetic deformation dataset since synthetic deformation data is generated with known point correspondence.

C. Synthetic Deformation Dataset Experiment (Part I)

In this section, experiments are conducted with the synthetic deformation dataset to show the superiority of the proposed method compared with the variants generated by combining different alternatives of the proposed two modules: spherical coordinate shape decomposition and angular deflections prediction as introduced in section II-B.

We first experiment with the methods that generate global Euclidean displacement as the deformation field. However, the results show that they do not achieve desired effects. When combining global Euclidean displacement with raw centerline inputs, the values of the smoothness preservation loss and the length preservation loss are relatively big with an unstable state during the training process. The deformed segments of the test samples reveal distinct jaggedness as shown in Fig.5. When combining global Euclidean displacement with decomposed centerline inputs, the training curve hardly converges. These phenomena might be caused by the inhomogeneity and large range of the global displacement field, making it difficult to be predicted and constrained. Therefore, the global deformation field is abandoned.

We further experiment with the methods that generate local deformation fields. Apart from the proposed method, there are five related variants in total (definitions are listed in Table II).



Fig. 5. The registration results of predicting global Euclidean displacement with raw data.

The network is tested every six epochs as the training process going on and the variations of the mPD values on the test set are shown in Fig.6. Table II shows the detailed mPD values after training convergence and the p-values of corresponding t-tests that evaluate if there are significant differences between the proposed method and other five variants. We can observe that the methods applying decomposed input achieve better performance than the ones applying raw data input. Besides, spherical coordinate decomposition is more effective than Euclidean coordinate decomposition. And angular deflection is more effective than Euclidean displacement when applied as deformation field. The visual registration results are presented in Fig.7. It can be concluded that the combination of spherical coordinate shape decomposition and angular deflections prediction has the best performance with an average error of 0.80 mm, which demonstrates the effectiveness of the proposed method. Detailed analysis is presented in section IV.

D. Synthetic Deformation Dataset Experiment (Part II)

The experiments in section III-C are conducted based on a general model that applies synthetic deformation dataset generated with all the commonly used projection angle settings as shown in Table I. However, a complete coronary artery structure consists of two main arteries: LCA and RCA with quite different appearances, which inspires us to establish artery-specific models only with the data of a certain artery. Besides, we can further extend this idea by establishing angle-specific models that only apply to the data under a fixed projection

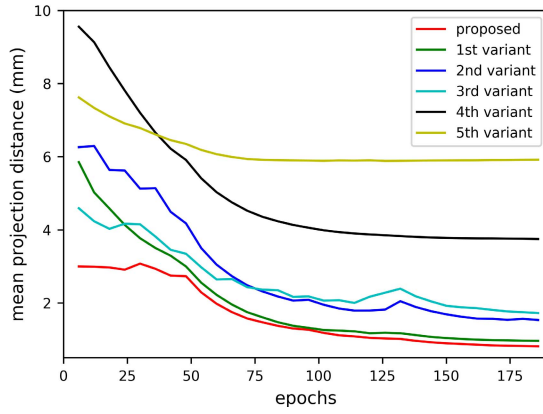


Fig. 6. The variation curves of mean projection distance (mm) of different methods on the test set. Related definitions can be referred to in Table II.

TABLE III

THE MEAN PROJECTION DISTANCE (MM) $\pm$ STANDARD DEVIATION OF THE GENERAL MODEL AND ARTERY-SPECIFIC MODELS AND THE P-VALUES OF CORRESPONDING T-TESTS

model	mPD (mm)	
	LCA	RCA
general	0.74 ± 0.49	0.87 ± 0.67
artery-specific	0.78 ± 0.51	0.82 ± 0.64
p-value	0.576	0.513

angle setting. We speculate that these specific models have the potential to achieve better performance since they do not involve extra variance caused by different coronary structures and projection angle settings. Therefore, experiments are conducted to show the performances of different models, and t-tests are also applied to evaluate if there are differences between the general model and specific models. Note that the general model in subsequent experiments is adopted from section III-C. The training set size of specific models is kept the same as that of the general model (4968). Corresponding test sets are generated by collecting eligible data from the general test set.

For establishing artery-specific models, 18 (LCA) and 36 (RCA) samples are generated from each patient in the clinical training set under each projection angle setting, respectively, to construct the synthetic deformation training set. The registration results on the test set and the p-values of corresponding t-tests are shown in Table III. We can observe that the general model seems to behave better when handling LCA and worse when handling RCA. However, the p-values indicate that there are no significant differences between the general model and artery-specific models.

Since the workload might be too heavy to experiment with all the nine groups of projection angle settings (α , β), three of them: (0, -30), (-30, 25) and (0, 25) are randomly selected to establish models. For each angle-specific model, the training set consists of 4968 coronary tree samples with 108 samples generated from each patient in the clinical training set. The registration results on the test set the p-values of corresponding

TABLE IV

THE MEAN PROJECTION DISTANCE (MM) $\pm$ STANDARD DEVIATION OF THE GENERAL MODEL AND ANGLE-SPECIFIC MODELS BASED ON DIFFERENT PROJECTION ANGLE SETTINGS (α , β) AND THE P-VALUES OF CORRESPONDING T-TESTS

model	mPD (mm)		
	(0, -30)	(-30, 25)	(0, 25)
general	0.74 ± 0.35	0.82 ± 0.84	0.76 ± 0.44
angle-specific	0.73 ± 0.35	0.82 ± 0.87	0.82 ± 0.39
p-value	0.794	0.947	0.351

t-tests are shown in Table IV. We can observe that the two types of models have similar performance under the setting of (0, -30) and (-30, 25). And the general model seems to have better performance than the angle-specific model under the setting of (0, 25). However, the p-values indicate that there are no significant differences between the general model and angle-specific models.

In summary, the experimental results demonstrate that establishing specific models is not essential since they do not contribute to actual registration performance improvement. This indicates that CAR-Net is robust to the type of coronary artery data, which will reduce the cost for clinical application since we only need to train a general model.

E. Clinical Dataset Experiment

Based on the general model, we further conduct experiments to evaluate the performance of the proposed CAR-Net on the clinical dataset.

As introduced before, the synthetic deformation dataset is established to increase data diversity and further improve the performance of the proposed CAR-Net on the clinical dataset. And the effectiveness of the synthetic deformation dataset is demonstrated by comparing the performances of two different neural network training strategies. The first strategy randomly initializes a raw network and trains from scratch with the clinical dataset. The second strategy adopts a two-stage training method, which first pre-trains the network with the synthetic deformation dataset and then fine-tunes it with the clinical dataset. Considering that the experiments in section III-C have indicated applying spherical coordinate shape decomposition and angular deflections prediction achieve the best performance, we continue to use these modules at this time. Note that when processing clinical data, the normalized soft-DTW distance is applied to measure the registration error in the loss function since it could align the projected CTA segment centerline with the target XCA segment centerline automatically during the training process. Fig.8 shows the variation curves of the soft-DTW training errors based on the two strategies. We can observe that fine-tuning a pre-trained model significantly accelerates convergence and produces a better solution. And when applying the obtained models to the clinical test set, an overall average error of 2.10 mm and 1.13 mm are achieved by training from scratch and fine-tuning a pre-trained model, respectively. The visual registration results of

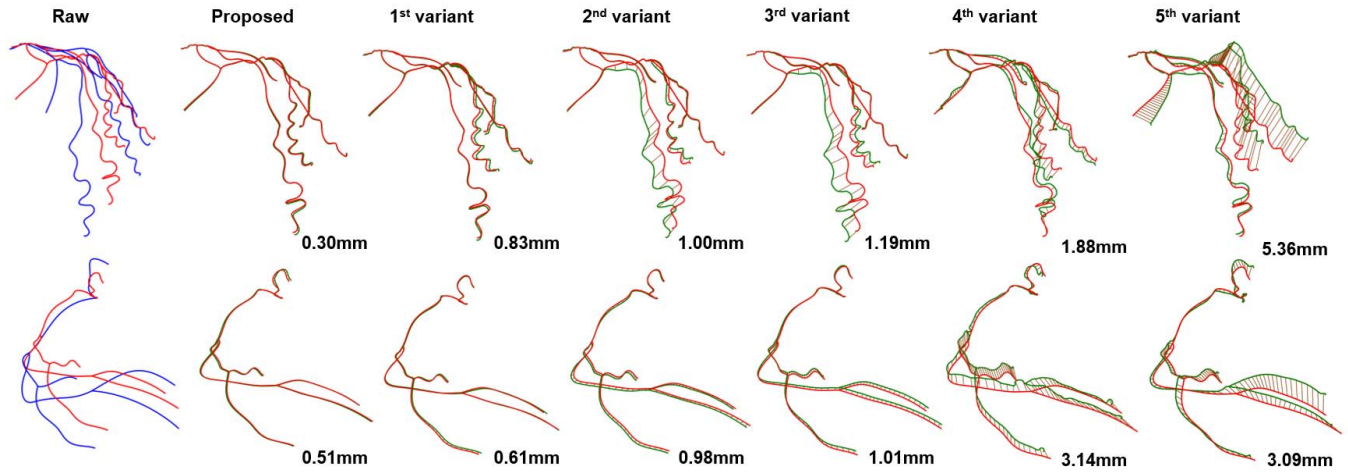


Fig. 7. The registration results of different methods on synthetic deformation data (first row: LCA sample; second row: RCA sample). Blue, red, and green curves denote the original 3D, the target 2D, and the registered 3D coronary structures, respectively. Brown short lines denote the gaps between the registration results and the targets. The registration errors are labeled at the right bottom.

TABLE V

THE MEAN PROJECTION DISTANCE (MM) $\pm$ STANDARD DEVIATION OF DIFFERENT METHODS ON CLINICAL TEST SET AND THE P-VALUES OF CORRESPONDING T-TESTS. IN INPUT, 'RAW': RAW DATA; 'SPHERICAL': SPHERICAL COORDINATE SHAPE DECOMPOSITION; 'EUCLIDEAN': EUCLIDEAN COORDINATE SHAPE DECOMPOSITION. IN DEFORMATION, 'EUCLIDEAN': LOCAL EUCLIDEAN DISPLACEMENT FIELD; 'ANGULAR': ANGULAR DEFLECTION FIELD

method	input			deformation		mPD (mm)	p-value
	spherical	Euclidean	raw	Euclidean	angular		
proposed	✓				✓	1.13 ± 0.83	/
1 st variant		✓			✓	1.44 ± 0.85	4.06e-02
2 nd variant	✓			✓		1.62 ± 0.74	1.81e-04
3 rd variant		✓		✓		1.56 ± 0.73	4.81e-04
4 th variant			✓		✓	3.02 ± 1.91	3.11e-12
5 th variant			✓	✓		3.30 ± 1.47	9.84e-15

some test samples based on the two training strategies are presented in the second row and third row of Fig.10 (a) and Fig.10 (b), respectively. We think that pre-training the network with synthetic deformation data could improve the robustness of CAR-Net to data variation. And subsequent fine-tuning makes the network adapt to real clinical data. Therefore, better registration performance is achieved compared with training from scratch.

In section III-C, we have conducted experiments with the synthetic deformation dataset to show the superiority of the proposed method which combines spherical coordinate shape decomposition and angular deflections prediction. And in this section, similar experiments are conducted to show that such superiority still exists for handling clinical data. Five variants are generated just as in section III-C. We fine-tune the pre-trained network models generated in section III-C with the clinical training set and evaluate their performances with the clinical test set. Table V shows the detailed mPD values of the six methods and p-values of corresponding t-tests. Fig. 9 shows the visual registration results of two samples based on the six methods. We can still get the conclusion that

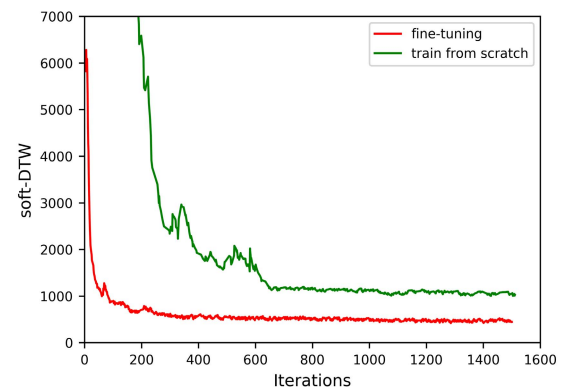


Fig. 8. The variation curves of the soft-DTW training errors based on fine-tuning a pre-trained model and training from scratch.

the proposed method achieves the best performance. Detailed analysis is presented in section IV.

We also try to compare the performance of the proposed CAR-Net with the deformation model TPS, which was widely applied in existing coronary artery registration

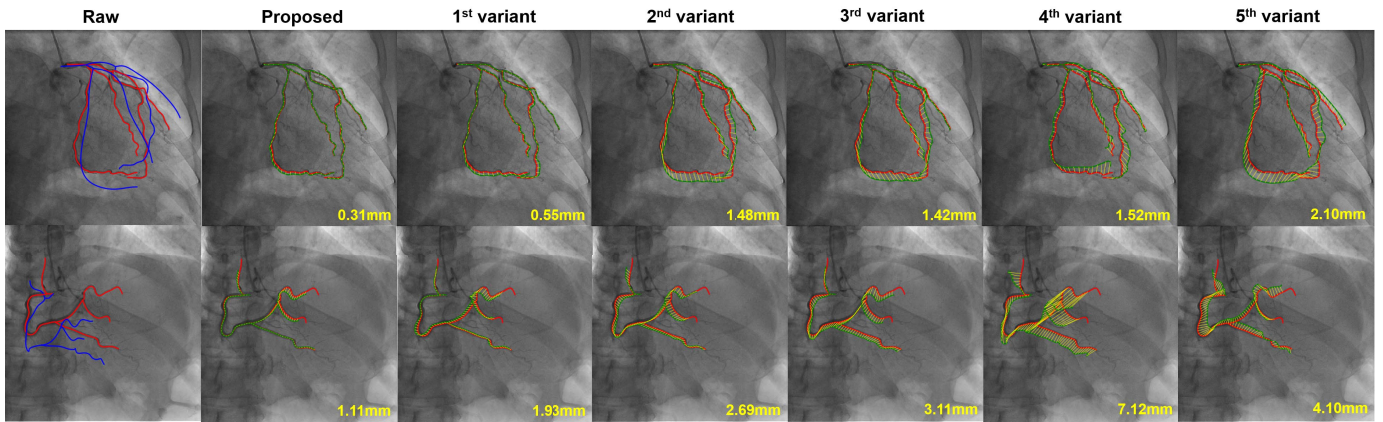


Fig. 9. The registration results of different methods on clinical data (first row: LCA sample; second row: RCA sample). Red, blue, and green curves denote the target 2D, the original 3D, and the registered 3D coronary artery structures, respectively. Yellow short lines denote the gaps between the registration results and the targets. The registration errors are labeled at the right bottom.

works [6], [7], [14]. And in our experiment, it's applied with Powell-Brent optimization method. TPS achieves an overall average error of 2.31 mm on our clinical test set. The visual registration results are presented in the fourth rows of Fig.10(a) and Fig.10(b). Considering that TPS requires iterative optimizations for each testing sample, the proposed CAR-Net is superior in both accuracy and efficiency.

In summary, by pre-training with the synthetic deformation dataset and adopting the proposed spherical coordinate shape decomposition and angular deflections prediction, CAR-Net could achieve an average error of 1.13 mm on real clinical data.

IV. DISCUSSION AND CONCLUSION

In this study, we propose a deep learning-based CAR-Net for Deformable 3D/2D coronary artery registration. The vessel segment centerlines which preserve topological information are decomposed into origin tensor and spherical coordinate shape tensor as the network input for intra-variance reduction. The network outputs the angular deflections as deformation fields, which implies motion and length preservation constraints for each vessel segment. The final registration result of a coronary tree is obtained by summarizing the result of each vessel segment, following the topological structure. The proposed method achieves an average registration error of 0.80 mm and 1.13 mm on the synthetic deformation dataset and clinical dataset, respectively. Although the performance is good, we believe there are still some contents worth discussing.

The robustness of the pre-processing steps might be the first problem that worthy noticing. An essential precondition for the segment-by-segment registration task is to find out the corresponding XCA vessel segment of each CTA vessel segment after vascular structure extraction, which first requires the segmentation algorithms to extract the vessel segments as completely as possible. In this study, the applied segmentation algorithms are based on neural networks and were developed by our lab and research partners recently. Manual interactions at the training stage and conditional random field were

adopted to make the extracted vascular structures continuous and smooth. Statistically, there are 2716 vessel segment pairs in the applied clinical dataset, the skeletons of which are obtained with morphological operations. And 89.2% of these vessel segment pairs are extracted completely. However, in certain cases, due to the low-does contrast agent or background interference, a thin vessel segment might be only partially segmented. This causes the difference of the vessel segment length between the two modalities, and the distal part of the longer vessel segment has no matching structure. Another worse situation is that the segmentation algorithm completely ignores the thin vessel segment and generates an incomplete vascular tree, which further makes the structure of the two modalities inconsistent. These two types of imperfect data take up 7.7% and 3.1% of the extracted vessel segment pairs, respectively. Although developing segmentation algorithms with higher performances might be a significant future work to solve this problem, currently we take some special strategies to handle these imperfect data since we mainly focus on establishing a deformation model in this study. For the first situation, considering the clinical value of the missing part is usually limited, the redundant distal part is removed to make the vessel segment length closer. For the second situation, the problem is handled by the applied matching algorithm that involves a tree topology consistency constraint. The vascular structure is reorganized by concatenating the preceding and neighboring vessels of the missing vessel, and registration is conducted without the missing vessel. After centerline extraction, the following vessel segment matching is another important step. And an overall matching accuracy of 0.916 is achieved. In summary, we think the pre-processing steps are generally robust and the influence of the imperfect results could be controlled.

Empirically, applying complete coronary artery centerlines as network input might be the most intuitive and ideal method since the registration results will be generated with only one forward propagation. However, due to the huge variations among the vessel structures of different patients, it's almost impossible to convert different coronary trees into a unified data structure while preserving the topological information

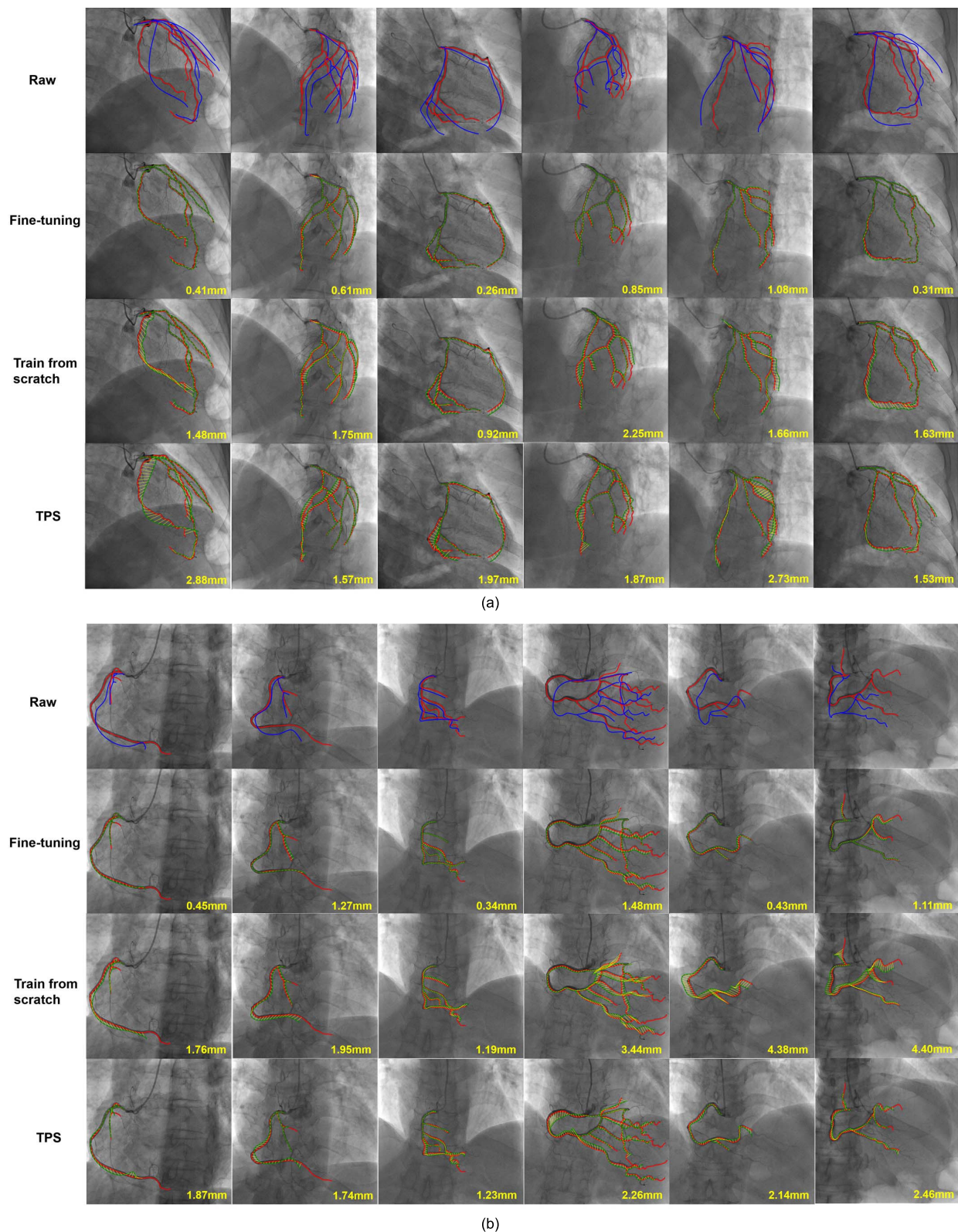


Fig. 10. The registration results of clinical data ((a): LCA; (b): RCA). From top to bottom, the four rows show the raw data before registration, the results of fine-tuning a synthetic deformation dataset pre-trained model, the results of training from scratch with the clinical data, and the results of tps, respectively. Red, blue, and green curves denote the target 2D, the original 3D, and the registered 3D coronary artery structures, respectively. Yellow short lines denote the gaps between the registration results and the targets. The registration errors are labeled at the right bottom.

that is essential for imposing constraints. Therefore, in this study, we adopt a segment-by-segment registration manner by interpolating each segment centerline into a uniform size. And

the overall registration result is obtained by combining the result of each vessel segment. However, to be objective, this manner also has some drawbacks. First, the efficiency will

be influenced by several forward propagations. Second, the origin of a vessel segment is determined by the endpoint of its preceding vessel segment, which will propagate the registration errors from upper to lower vessel segments, causing error accumulation.

After the segment-by-segment manner is determined, we conduct the experiment by first applying the raw centerline data as input since this is intuitive. However, we found that the running mean and variance of the batch normalization modules in convolutional layers are in an unstable state during the training process, resulting in non-convergence. This is because the input centerline points have a relatively wide distribution space with variable coordinate values. The large gap between the mean and variance of different input data batches damages the training process. This issue can be addressed to some extent by replacing batch normalization with instance normalization. Now the training curve converges, however, the performance of the algorithm is not satisfying. Centerline decomposition is proposed to solve this problem by reducing the intra-variance of the input data. The position of a centerline curve is decided by its origin and shape. Since a curve is densely sampled into n points, the variance of the shape data is relatively small. The high variance among the coordinate values is decided by the origin points. Therefore, by decomposing the centerline data and processing the origin and shape data with different branches, the influence of the high variance is diminished and restricted to the origin branch. And after information fusion, we can now observe that the running mean and variance reveal a steady state. Centerline decomposition achieves a faster convergence speed and produces a better solution when predicting the local deformation field.

Based on centerline decomposition, we further conduct experiments to find the most effective decomposition method. The results on both the synthetic deformation dataset and the clinical dataset demonstrate that the proposed method, which combines spherical coordinate shape decomposition and angular deflections prediction, achieve the best performance. This indicates the effectiveness of modeling the shape and deformation of curvilinear vessel structures with a spherical coordinate system, which we believe has at least three advantages. First, the absolute motion of each point of the centerline is determined by the relative motions of all its preceding points, which implies broader motion constraints. Second, by fixing l and varying θ and ϕ , an implicit constraint for the vessel length is imposed. We find that when predicting Euclidean displacement, although an explicit length constraint is added, there is still a length variation of about 8% before and after registration, which is unacceptable. Note that although the vessels might stretch in certain cases, the inextensibility is locally correct since the centerline points are densely sampled. This highlights the importance of applying angular deflections as the deformation field for vessel length preservation. Third, there are only two parameters $\Delta\theta$ and $\Delta\phi$ require predicting for each point, while the existing methods usually predicted three parameters: the Euclidean displacements in the X, Y, and Z-axis. Considering that there are substantial points, this could greatly reduce the number of parameters.

Finally, we also want to compare the proposed method with existing algorithms. However, most of the previous literature is published years ago with no open-source codes, which makes it difficult to reproduce related results. Therefore, in section III-E, a comparative experiment is conducted just between the proposed CAR-Net and the commonly applied deformation model TPS. Results show that CAR-Net is superior in both accuracy and efficiency. And compared with the latest coronary artery registration work [16] which reports errors of 1.15 mm on LCA and 1.59 mm on RCA, the proposed CAR-Net is also competitive. In summary, we believe the proposed CAR-Net has the potential to be used in clinical practice. And in the future, we will improve this study from two aspects. First, we will cooperate with more hospitals to collect clinical data with more variations, establishing a larger dataset. Second, we will try to extend the proposed method to the vascular structures of other organs since some parts of the method are general.

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